

N.B.: (1) All questions are compulsory.

(2) From Questions 1, 2 and 3. Attempt any one from part(a) and any two from part(b).

(3) From Question 4. Attempt any three.

(4) Figures to the right indicate marks.

1. (a) (i) If G is self complementary graph of order p , show that G is connected and $p \equiv 0$ or $1 \pmod{4}$ (8)
- (ii) If G is simple graph with p vertices, q edges and k components, then prove that $q \geq p - k$ and hence prove that a connected graph of order p has at least $p - 1$ edges. (8)
- (b) (i) Define a cut vertex and cut edge of a graph. Let G be a connected graph with at least 3 vertices and if G contains a cut edge then prove that it contains a cut vertex. (6)
- (ii) State Havel-Hakimi theorem for degree sequence of a graph. Using the theorem check whether the sequence $5, 4, 3, 3, 2, 2, 2, 1$ is graphical and if so, draw the graph with this degree sequence. (6)
- (iii) If G is simple graph on n vertices and $\delta(G) \geq \frac{n-1}{2}$ then prove that G is a connected graph where $\delta(G)$ denotes the minimum degree of G . Give an example of a graph with $\delta(G) \geq \frac{n-1}{2}$ which is not connected. (6)
- (iv) Show that a connected (p, q) graph contains a cycle if and only if $q \geq p$ (6)
2. (a) (i) State and prove Cayley's formula for spanning trees. (8)
- (ii) Let G be (p, q) graph. Show that the following statements are equivalent. (8)
 - 1) G is tree.
 - 2) G is acyclic and $q = p - 1$.
 - 3) G is connected and $q = p - 1$.
- (b) (i) Define a binary tree. Show that the number of leaves in a binary tree with n vertices is $\frac{n+1}{2}$ (6)
- (ii) Prove that if G is a nontrivial tree then there exists at least 2 vertices which are pendant vertices. (6)
- (iii) If T is tree with n vertices whose degree sequences is $d_1, d_2, \dots, d_n$, then prove that $\sum_{i=1}^n d_i = 2(n-1)$ (6)
- (iv) Show that a graph is connected if and only if it has a spanning tree. (6)
3. (a) (i) Show that the cube graph Q_k is connected bipartite k -regular graph with 2^k vertices. (8)
- (ii) If G is a graph on p vertices with $p \geq 3$ such that $\deg(u) + \deg(v) \geq p$ for every pair of non adjacent vertices u and v in G , then prove that G is Hamiltonian. (8)
- (b) (i) Define closure of a graph $C(G)$ and show that if the closure of (6)

- graph G is complete then G is Hamiltonian.
- (ii) Let G be a simple graph with p vertices and q edges with $p \geq 3$. If $q \geq \frac{p^2-3p+6}{2}$ then prove that G is Hamiltonian. (6)
- (iii) Show that there are n edge disjoint Hamiltonian cycles in K_{2n+1} . (6)
- Also draw all edge disjoint Hamiltonian cycles in K_7 .
- (iv) If G is a graph on p vertices with $\deg(u) + \deg(v) \geq p-1$ for every pair of non adjacent vertices u and v in G , then show that G contains a Hamiltonian path. (6)
- (i) Let A denote the adjacency matrix of a connected graph G with $V(G) = \{v_1, v_2, \dots, v_n\}$, then show that the distance between v_i and v_j is the smallest integer $n \geq 0$ such that $(A^n)_{ij} \neq 0$ (5)
- (ii) Show that a vertex v is a cut vertex if and only if there exists two vertices x and y such that v is on every $x - y$ path in G . (5)
- (iii) Let $\tau(G)$ denote the number of spanning trees of a graph G . If $e \in E(G)$ is not a loop then prove that $\tau(G) = \tau(G - e) + \tau(G \cdot e)$ (5)
- (iv) Prove that vertex connectivity of a graph is less than or equal to edge connectivity of the graph. (5)
- (v) If G is Hamiltonian graph then for every nonempty proper subset S of $V(G)$, prove that $\omega(G - S) \leq |S|$. (5)
- (vi) Prove that the cube graph Q_n , with $n \geq 2$ is a Hamiltonian graph. (5)

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